Genel olarak iyi ama çok fazla anahtar sözcük kullanılmış. Ek olarak bir yapay zeka aracıyla yazılmış olabileceğiyle ilgili bir izlenim edindim.

UAV Indoor Exploration with Visual SLAM Using Hybrid Frontier and Information Gain Planning

Batuhan ODÇIKIN
Bilgisayar Mühendisliği Bölümü
Yıldız Teknik Üniversitesi, 34220 Istanbul, Türkiye
{batuhan.odcikin}@std.yildiz.edu.tr

***Özetçe* —Bu calismada, insansiz hava araci icin kapali ve dar alanlarda gercek dunyaya uyarlanabilir otonom kesif yapabilen hibrit bir yaklasim gelistirilmistir. Sistem, frontier tabanli kesfi bilgi kazanimi temelli yontemlerle birlestirerek verimli bir ic ortam kesfi amaclar. ROS 2 altyapisinda moduler bir mimari ile tasarlanan cozum, Gazebo simulasyonunda PX4 SITL kullanilarak test edilmistir. Derinlik kamerasi nokta bulutu verisi OctoMap ile uc boyutlu voksel haritaya donusturulmus, planlama ise bu haritanin iki boyutlu kesiti uzerinden gerceklestirilmistir. Algoritma; BFS tabanli kumeleme ve PCA ile sinir bolgelerinin tespiti, silindirik ornekleme ve engelleme duyarli kapsama analizi ile bakis noktalari uretilmesi, mesafe kume buyuklugu acisal degisim ve kapsama olcutlerini iceren cok amacli maliyet fonksiyonu ile hedef secimi, Nav2 navigasyon yigini ve yonelim enjeksiyonu ile yol planlama ve takip asamalarindan olusur. Sistem, farkli zorluk seviyelerine sahip uc ayri ortamda degerlendirilmis ve uc bin iki yuz metrekareye varan alanlarda yuzde doksan besin uzerinde kapsama oranina ulasirken, ortalama bir nokta iki metre saniye hiz mertebesinde ilerleyip sifir nokta otuz uc yuzde duzeyinde dusuk durma orani ile akici kesif davranisi sergilemistir.**

***Anahtar Kelimeler*—kapali alan kesfi, insansiz hava araci, otonom kesif, gorsel slam, bilgi kazanimi, frontier tabanli kesif, OctoMap, bakis noktasi planlama, yol planlama, ROS 2, Gazebo, PX4.**

***Abstract*—In this study, a hybrid algorithm for unmanned aerial vehicles is developed to enable autonomous exploration in enclosed and confined indoor environments with a focus on real-world adaptability. The proposed system combines frontier-based exploration with information-gain methods to improve exploration efficiency. It is implemented as a modular architecture under the ROS 2 framework and evaluated in the Gazebo simulation environment using PX4 SITL. Point cloud data from a depth camera is converted into a three-dimensional voxel map with OctoMap, while exploration planning is performed on a two-dimensional cross section derived from this map. The algorithm comprises four stages: boundary region detection via BFS-based clustering and PCA partitioning, viewpoint generation using cylindrical sampling with obstruction-sensitive coverage analysis, target selection through a multi-objective cost function that includes distance, cluster size, angular change and coverage, and path planning and tracking using the Nav2 navigation stack with an orientation injection technique. Experiments in three indoor scenarios with different difficulty levels demonstrate that the system achieves over ninety-five percent coverage in areas up to three thousand two hundred square meters, while maintaining smooth motion at around one point two meters per second and a low stopping rate of zero point three three percent.**

***Keywords*—indoor exploration, unmanned aerial vehicle, autonomous exploration, visual SLAM, information gain, frontier-based exploration, OctoMap, viewpoint planning, path planning, trajectory tracking, ROS 2, Gazebo simulation, PX4 SITL.**

I. Introduction

Unmanned Aerial Vehicles (UAVs) are increasingly adopted for exploration, inspection, and search-and-rescue missions, particularly in hazardous or unknown environments such as partially collapsed buildings, underground facilities, and narrow indoor corridors. In these settings, reliable autonomy is challenging because GPS is not available and perception is strongly affected by occlusions, repetitive textures, and limited field-of-view sensing. Consequently, an autonomous indoor exploration pipeline must jointly address three coupled problems: (i) localization under perceptual uncertainty, (ii) mapping of free/occupied/unknown space, and (iii) action selection that balances safety, efficiency, and coverage.

This study focuses on a simulation-based UAV exploration system where pose estimation is provided by Visual SLAM (VSLAM) supported by inertial measurements and depth sensing. The main objective is to design a planning approach that is efficient in confined indoor settings while remaining practical for real-time execution. Rather than using a purely greedy frontier selection, the proposed method combines frontier-based exploration with information-gain-aware viewpoint evaluation, aiming to reduce unnecessary revisits and to maintain smoother navigation behavior.

The key contributions of this work can be summarized as follows. First, a modular ROS 2-based software architecture is presented, integrating Gazebo simulation, PX4 offboard control, OctoMap-based mapping, and Nav2-based navigation. Second, a hybrid exploration algorithm is implemented that detects and clusters frontier regions, generates candidate viewpoints through cylindrical sampling, evaluates them via obstruction-aware coverage estimation, and selects targets with a multi-objective cost that accounts for distance, cluster size, yaw change and expected gain. Finally, the overall system is evaluated in three indoor scenarios with increasing difficulty, and its coverage and motion smoothness are reported.

II. Literature Review

Autonomous exploration can be framed as answering the question: “Where should the robot go next?” The classical literature groups exploration strategies into frontier-based methods, information-gain-driven viewpoint methods, and hybrid methods that combine both.

A. Frontier-Based Exploration

Frontier-based exploration was introduced by Yamauchi [1]. In this paradigm, frontiers are defined as boundaries between known free space and unknown space. A robot repeatedly detects frontier cells in its current map and selects one as a navigation goal. The method is appealing due to its conceptual simplicity and low computational overhead, which makes it suitable for real-time deployment. However, in cluttered indoor environments, purely greedy frontier selection can lead to myopic behavior and frequent revisits, especially when narrow passages and dead ends are present. This may increase overall travel distance and reduce global exploration efficiency.

B. Information-Gain-Based Exploration

Information-gain approaches estimate how much new information a candidate viewpoint is expected to reveal. Connolly proposed the classic "next best view" formulation [2], where candidate viewpoints are evaluated based on their expected utility. For aerial robots exploring 3D environments, the planning problem becomes significantly harder due to high-dimensional state spaces and the need for collision-free trajectories. Receding-horizon NBV planning [3] addresses this challenge by limiting the planning horizon and repeatedly replanning, enabling efficient exploration in unknown 3D spaces. Despite strong performance, information-gain planning can incur high computational costs due to repeated viewpoint sampling, ray casting, and path evaluation.

C. Hybrid Exploration

Hybrid approaches aim to preserve the speed of frontier exploration while benefiting from information-gain-driven decision making. FUEL [4] proposes incremental frontier structures and hierarchical planning to speed up target selection and trajectory computation. FAEP [5] focuses on fast exploration under limited sensor field-of-view constraints. ECHO [6] introduces efficient heuristic viewpoint determination to reduce computation while maintaining exploration quality. FALCON [7] leverages coverage-path guidance to improve global exploration consistency and reduce redundant revisits. These works motivate a design that uses frontiers as exploration cues but selects targets based on a gain-aware scoring mechanism, particularly important for indoor UAV exploration where maneuverability and stability constraints play a major role.

III. System Analysis and Feasibility

A. System Architecture

The system is implemented in ROS 2 and consists of three main components: Gazebo Harmonic as the simulation environment, PX4 Autopilot in software-in-the-loop (SITL) mode for flight control, and a custom exploration module implemented as ROS 2 nodes. PX4 provides a node-based multithreaded control framework and is widely used in aerial robotics research and development [8]. In the proposed design, Gazebo simulates physics and generates sensor outputs such as RGB-D point clouds and IMU readings. The exploration module performs mapping, frontier detection, target selection and navigation planning. Selected targets are converted into offboard commands and sent to PX4, which produces motor-level control signals. ROS 2 pub/sub communication provides real-time feedback loops and enables modular integration of perception, mapping and planning.

B. Real-Time Considerations

Indoor exploration requires synchronization across perception, mapping, planning and control. Depth sensing is typically produced at video rate, while inertial data is available at higher frequency. A practical exploration pipeline must therefore tolerate sensor latency and variable computation times while maintaining stable control. In this work, the software design separates mapping updates and goal selection from low-level control. The UAV continuously receives offboard setpoints, while the exploration planner updates targets at a slower rate to avoid oscillations. This separation is particularly important for frontier-based exploration, because frequent goal switches can destabilize the vehicle and lead to excessive stopping.

C. Evaluation Metrics

Performance is evaluated by measuring exploration time to reach a target coverage level, final coverage ratio, coverage rate, total traveled distance, average flight speed, and stopping rate as an indicator of motion smoothness. These metrics jointly capture both exploration effectiveness (coverage) and execution quality (smoothness and efficiency). The stopping rate is especially relevant for confined indoor flight because frequent stopping implies excessive replanning or unstable target switching, which can increase mission time and energy consumption.

D. Feasibility

The proposed approach is fully simulation-based, which allows extensive experimentation without safety risks or hardware limitations. The selected software stack (Ubuntu 22.04, ROS 2 Humble, Gazebo Harmonic, PX4 SITL) is widely adopted and provides the required interfaces for migrating the system to real hardware in future work. In particular, the modular ROS 2 design supports replacing simulated sensor sources with real drivers, and the offboard control interface can be reused with minimal changes.

IV. System Design

A. Mapping and Representation

The exploration pipeline relies on an occupancy representation derived from depth point clouds. The system constructs a probabilistic 3D voxel map using OctoMap [9], which provides a compact and flexible representation suitable for incremental updates. While 3D mapping is necessary for aerial robots, running full 3D exploration planning can be computationally expensive in real time.

Therefore, planning decisions are computed on a 2D cross section derived from the 3D voxel map, which reduces computational cost while preserving obstacle structures relevant for indoor navigation. This design choice is motivated by typical indoor environments where most navigation decisions are driven by horizontal layout and narrow passages, and it enables faster frontier extraction and scoring.

B. Frontier Extraction and Clustering

Frontiers are detected as boundary regions between known free space and unknown space on the planning layer. Detected frontier cells are grouped using BFS-based clustering to form coherent frontier segments. Clustering reduces noise and prevents the planner from reacting to isolated frontier pixels. To further characterize frontier geometry, PCA-based partitioning is applied to estimate dominant directions and to support robust viewpoint generation. This is useful in corridors and narrow openings where frontier geometry is elongated; in such cases, viewpoint candidates must be placed to maximize visibility without violating collision constraints.

C. Viewpoint Generation with Information Gain

For each frontier cluster, candidate viewpoints are generated through cylindrical sampling around the frontier region. The sampling strategy places candidates at feasible radii and angles relative to the cluster and proposes a set of yaw orientations that align with the expected observation direction. Each candidate is evaluated using obstruction-sensitive coverage analysis. Conceptually, the evaluation estimates how much unknown area becomes observable from the candidate viewpoint, which serves as an information-gain proxy similar in spirit to NBV formulations [2]. Compared to purely selecting the closest frontier, gain-aware evaluation reduces revisits because targets that reveal larger unknown regions are preferred.

D. Target Selection via Multi-Objective Cost

After viewpoint generation, the system selects a target using a multi-objective cost that balances exploration utility and motion quality. The cost includes travel distance (or a time proxy), frontier cluster size (favoring larger unexplored regions), yaw change (penalizing aggressive heading rotations), and expected coverage/gain. This design aligns with the motivation of hybrid planners such as FUEL [4] and FALCON [7], where efficient exploration requires both global progress and smooth execution. Penalizing yaw change is particularly important for UAVs because aggressive yawing can increase controller effort and may degrade VSLAM quality if visual features become unstable.

E. Navigation, Tracking, and Orientation Handling

Path planning and tracking are performed using the Nav2 navigation stack. While Nav2 is traditionally associated with ground robots, its modular planning and control pipeline is useful for waypoint-level navigation and local trajectory tracking. In this system, the planner produces collision-free paths on the derived planning layer, and path following is achieved using regulated pure pursuit concepts for stable tracking behavior [10]. To ensure that the UAV observes frontiers effectively, an “orientation injection” mechanism is applied: the yaw command is chosen to align the vehicle with the selected viewpoint’s intended observation direction. This reduces the mismatch between where the UAV moves and where it looks, which is important for gain-driven exploration.

V. Experimental Setup and Results

A. Simulation Environments

The exploration system is evaluated in three indoor scenarios with increasing difficulty. The “Complex Office” environment represents typical office-like structures with rooms and corridors. “Octamaze” contains narrow passages and frequent turns, stressing both frontier extraction and safe navigation. The “DARPA-style indoor course” resembles real-world constrained indoor layouts and is used to test robustness under occlusions and limited visibility. These environments were selected to cover both open-space exploration and highly constrained maze-like exploration where greedy frontier strategies often perform poorly [1].

B. Experimental Protocol

Each experiment starts with the UAV at a predefined initial pose in the environment. The UAV performs autonomous exploration until a target coverage threshold is reached or until the environment is effectively exhausted. The planner operates in a loop, repeatedly updating the map, extracting and clustering frontiers, generating candidate viewpoints, computing the multi-objective target score, and sending the resulting goal to the navigation system. During execution, the controller receives continuous offboard setpoints to maintain stable flight while moving along the planned path. Metrics are logged throughout the mission to quantify both exploration performance and motion smoothness.

C. Results and Discussion

Across environments up to approximately 3200 m^2, the system achieves over 95% coverage while maintaining smooth motion at around 1.2 m/s average speed. The stopping rate remains low (approximately 0.33%), indicating that the planner avoids excessive goal switching and provides stable exploration behavior. The results suggest that combining frontier cues with gain-aware viewpoint selection improves exploration efficiency compared to naive frontier chasing, particularly in maze-like indoor layouts where the closest frontier may lead to dead ends and repeated revisits.

From a qualitative perspective, the hybrid scoring mechanism helps prioritize frontier clusters that represent larger unknown regions, while the yaw change penalty and orientation injection improve execution smoothness. This behavior is consistent with the motivation behind hybrid exploration systems such as FUEL [4], FAEP [5], ECHO [6], and FALCON [7], which emphasize fast decision making while preserving global exploration progress. Moreover, the use of OctoMap [9] provides a reliable 3D occupancy backbone, while the 2D planning cross section improves real-time feasibility by limiting planning complexity.

D. Limitations

Despite promising results, several limitations remain. First, planning decisions are computed on a 2D cross section, which may be insufficient in multi-level indoor structures or when vertical exploration is required. Second, information gain is estimated using an obstruction-sensitive coverage proxy, which may deviate from true entropy-based gain in highly cluttered environments. Third, as in receding-horizon planning approaches [3], frequent replanning can still introduce computational load if viewpoint sampling is too dense. These limitations motivate improvements in gain estimation and hierarchical 3D decision making.

VI. Conclusion

This work presents a ROS 2-based autonomous indoor exploration framework for UAVs that combines frontier-driven cues with information-gain-aware viewpoint selection. The system integrates PX4 SITL control [8], OctoMap-based 3D mapping [9], and Nav2-based planning and tracking with regulated pure pursuit concepts [10]. The hybrid cost function incorporates distance, cluster size, yaw change, and estimated gain to balance exploration efficiency with smooth execution. Experiments in three indoor scenarios indicate that the system can achieve over 95% coverage in environments up to 3200 m^2 while maintaining approximately 1.2 m/s average speed and a low stopping rate.

Future work will focus on extending the planner to full 3D exploration, improving gain estimation under occlusion, and evaluating the approach on real hardware. In addition, incorporating receding-horizon 3D planning ideas [3] may further improve robustness in complex indoor environments while preserving real-time feasibility.

References

[1] B. Yamauchi, "A frontier-based approach for autonomous exploration," in *Proceedings 1997 IEEE International Symposium on Computational Intelligence in Robotics and Automation CIRA'97. 'Towards New Computational Principles for Robotics and Automation'*, 1997, pp. 146–151.

[2] C. Connolly, "The determination of next best views," in *Proceedings. 1985 IEEE International Conference on Robotics and Automation*, vol. 2, 1985, pp. 432–435.

[3] A. Bircher, M. Kamel, K. Alexis, H. Oleynikova, and R. Siegwart, "Receding horizon "next-best-view" planner for 3d exploration," in *2016 IEEE International Conference on Robotics and Automation (ICRA)*, 2016, pp. 1462–1468.

[4] B. Zhou, Y. Zhang, X. Chen, and S. Shen, "Fuel: Fast uav exploration using incremental frontier structure and hierarchical planning," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 779–786, 2021.

[5] Y. Zhao, L. Yan, Y. Chen, H. Xie, and B. Xu, "Faep: Fast autonomous exploration planner for uav equipped with limited fov sensor," 2022. [Online]. Available: https://arxiv.org/abs/2202.12507

[6] J. Yu, H. Shen, J. Xu, and T. Zhang, "Echo: An efficient heuristic viewpoint determination method on frontier-based autonomous exploration for quadrotors," *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 5047–5054, 2023.

[7] Y. Zhang, X. Chen, C. Feng, B. Zhou, and S. Shen, "Falcon: Fast autonomous aerial exploration using coverage path guidance," *IEEE Transactions on Robotics*, vol. 41, pp. 1365–1385, 2025.

[8] L. Meier, D. Honegger, and M. Pollefeys, "Px4: A node-based multithreaded open source robotics framework for deeply embedded platforms," in *2015 IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 6235–6240.

[9] K. M. Wurm, A. Hornung, M. Bennewitz, C. Stachniss, and W. Burgard, "Octomap: A probabilistic, flexible, and compact 3d map representation for robotic systems," in *Proc. of the ICRA 2010 workshop on best practice in 3D perception and modeling for mobile manipulation*, vol. 2, 2010, p. 3.

[10] S. Macenski, S. Singh, F. Martín, and J. Ginés, "Regulated pure pursuit for robot path tracking," *Autonomous Robots*, vol. 47, no. 6, pp. 685–694, 2023. [Online]. Available: https://doi.org/10.1007/s10514-023-10097-6